Report on Bhuvan and SRTM Data

1. Introduction

This report explores geospatial data available from Bhuvan and SRTM.

These platforms provide satellite imagery and elevation data that are useful for environmental analysis and Artificial Intelligence (AI) applications such as pattern recognition.

2. Bhuvan Exploration

Bhuvan is an Indian geoportal developed by ISRO that provides access to satellite imagery and thematic datasets related to natural resources and environmental monitoring.

The platform allows users to visualize data through interactive maps by applying different layers such as land use, water resources, vegetation, and soil.

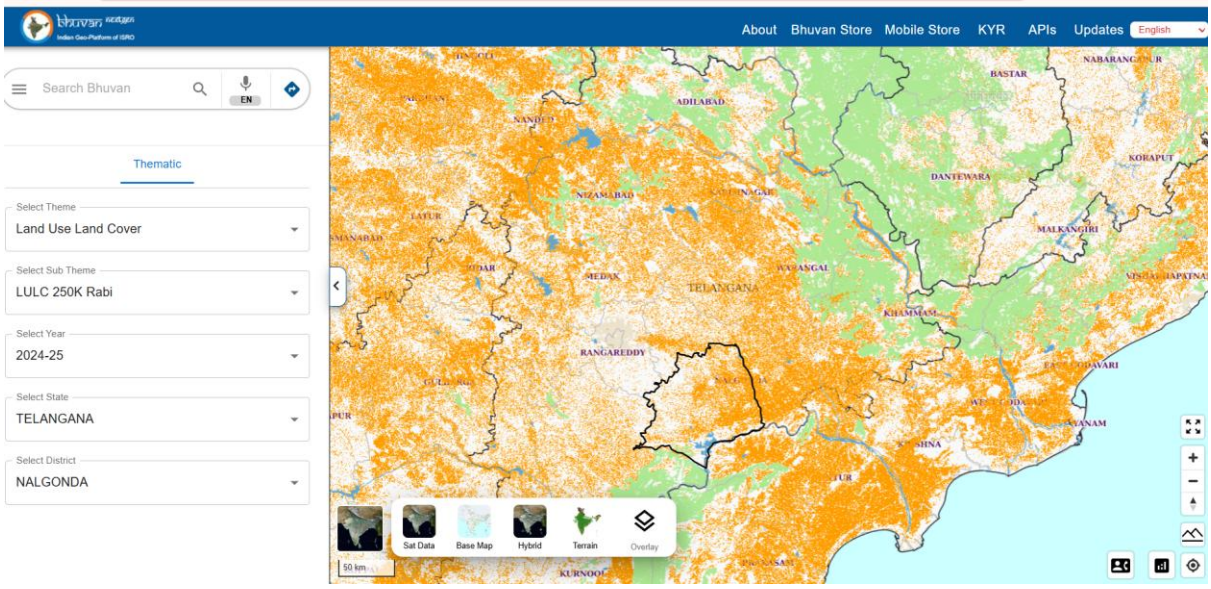


Figure1: Bhuvan Map with Layer Applied

3. Data Availability in Bhuvan

Bhuvan provides various types of geospatial datasets, including:

- Vegetation Index (NDVI)
- Water bodies and water resources data
- Soil moisture datasets
- Atmospheric and environmental data

These datasets are available with different spatial resolutions ranging from meters to kilometers and are updated at different time intervals such as daily, monthly, or seasonal.

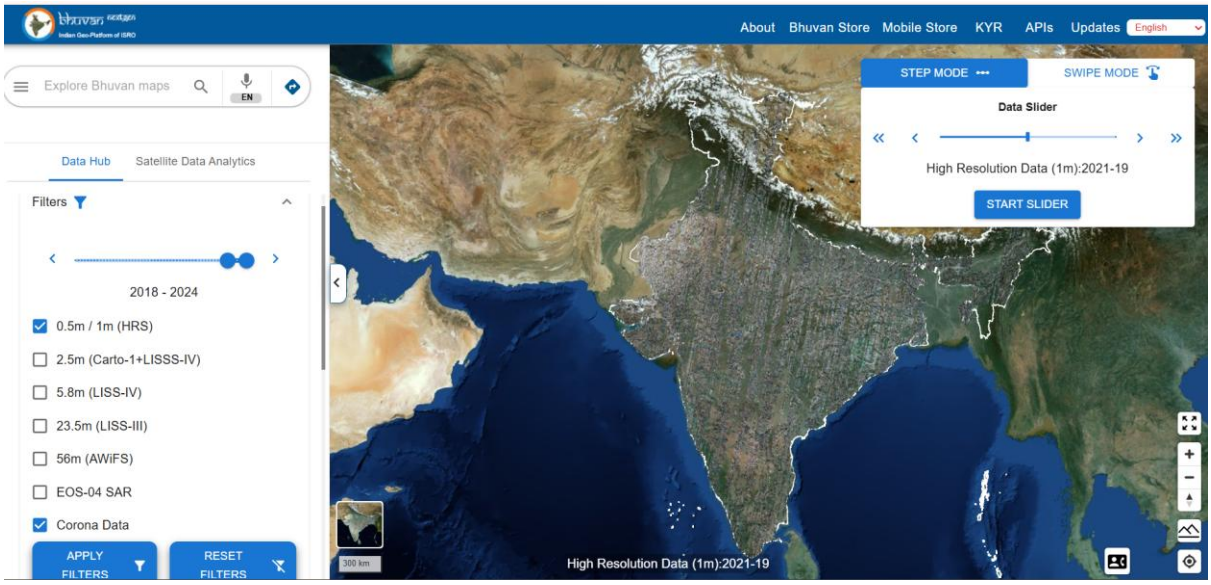
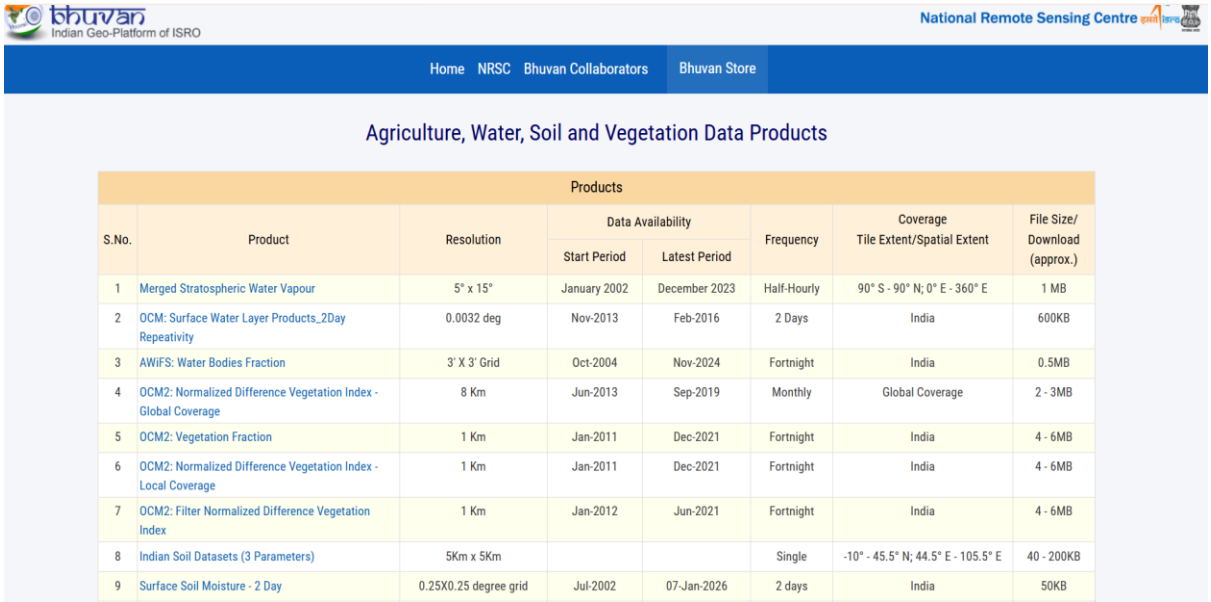


Figure 2: Data Hub / Filters in Bhuvan



Agriculture, Water, Soil and Vegetation Data Products

Products							
S.No.	Product	Resolution	Data Availability		Frequency	Coverage Tile Extent/Spatial Extent	File Size/ Download (approx.)
			Start Period	Latest Period			
1	Merged Stratospheric Water Vapour	5° x 15°	January 2002	December 2023	Half-Hourly	90° S - 90° N; 0° E - 360° E	1 MB
2	OCM: Surface Water Layer Products_2Day Repeativity	0.0032 deg	Nov-2013	Feb-2016	2 Days	India	600KB
3	AWiFS: Water Bodies Fraction	3' X 3' Grid	Oct-2004	Nov-2024	Fortnight	India	0.5MB
4	OCM2: Normalized Difference Vegetation Index - Global Coverage	8 Km	Jun-2013	Sep-2019	Monthly	Global Coverage	2 - 3MB
5	OCM2: Vegetation Fraction	1 Km	Jan-2011	Dec-2021	Fortnight	India	4 - 6MB
6	OCM2: Normalized Difference Vegetation Index - Local Coverage	1 Km	Jan-2011	Dec-2021	Fortnight	India	4 - 6MB
7	OCM2: Filter Normalized Difference Vegetation Index	1 Km	Jan-2012	Jun-2021	Fortnight	India	4 - 6MB
8	Indian Soil Datasets (3 Parameters)	5Km x 5Km			Single	-10° - 45.5° N; 44.5° E - 105.5° E	40 - 200KB
9	Surface Soil Moisture - 2 Day	0.25X0.25 degree grid	Jul-2002	07-Jan-2026	2 days	India	50KB

Figure 3: Dataset/Product Table (NDVI, Water, Soil Data)

4. Data Organization

The data in Bhuvan is not stored as individual images. Instead, it is organized in the form of:

- Thematic datasets
- Region-based tiles
- Time-series data

This structure helps in efficient storage, retrieval, and analysis of large-scale geospatial information.

5. Downloaded Data

From Bhuvan, the following data files were downloaded:

- **NetCDF file (.nc)** – used for storing scientific and multidimensional data
- **PNG file (.png)** – used for visual representation of spatial data

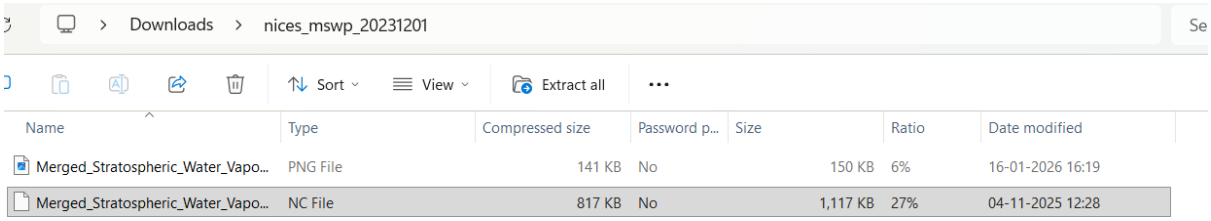


Figure 4: Downloaded Files (.nc and .png)

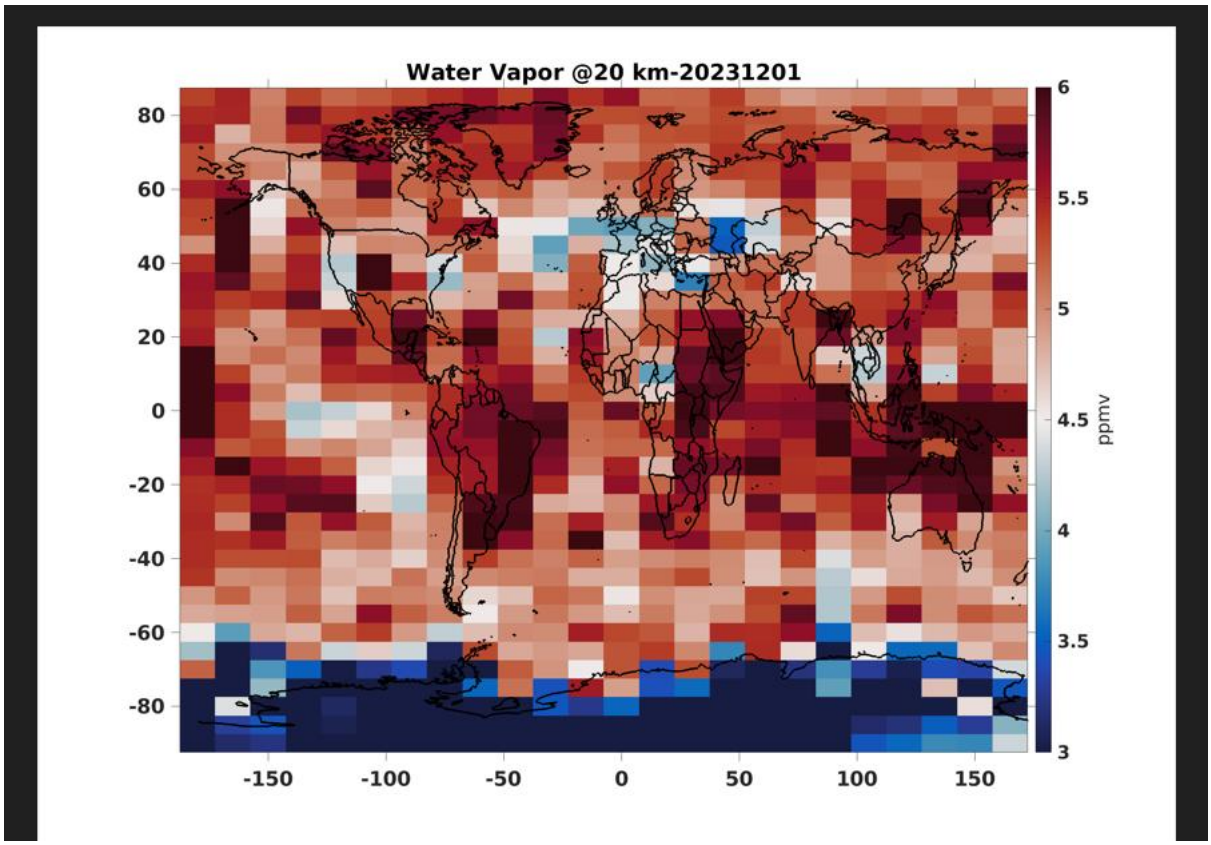


Figure 5: PNG Image Preview

Observations

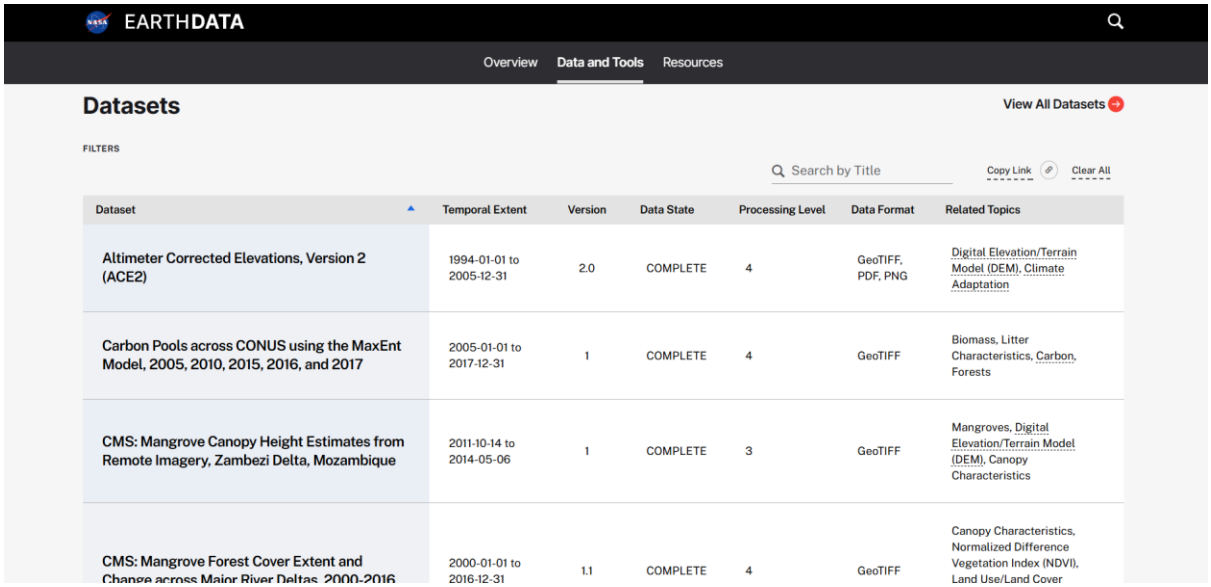
- The PNG image shows the spatial distribution of features such as vegetation or water bodies.

- The NetCDF file contains structured geospatial data that can be used for advanced analysis and machine learning applications.

6. SRTM Data

SRTM provides global elevation data collected using radar systems.

- Source: USGS Earth Explorer
- Resolution: 30 meters
- Coverage: Global (including India)
- Data Type: Digital Elevation Model (DEM)



Dataset	Temporal Extent	Version	Data State	Processing Level	Data Format	Related Topics
Altimeter Corrected Elevations, Version 2 (ACE2)	1994-01-01 to 2005-12-31	2.0	COMPLETE	4	GeoTIFF, PDF, PNG	Digital Elevation/Terrain Model (DEM), Climate Adaptation
Carbon Pools across CONUS using the MaxEnt Model, 2005, 2010, 2015, 2016, and 2017	2005-01-01 to 2017-12-31	1	COMPLETE	4	GeoTIFF	Biomass, Litter Characteristics, Carbon, Forests
CMS: Mangrove Canopy Height Estimates from Remote Imagery, Zambezi Delta, Mozambique	2011-10-14 to 2014-05-06	1	COMPLETE	3	GeoTIFF	Mangroves, Digital Elevation/Terrain Model (DEM), Canopy Characteristics
CMS: Mangrove Forest Cover Extent and Change across Major River Deltas, 2000-2016	2000-01-01 to 2016-12-31	1.1	COMPLETE	4	GeoTIFF	Canopy Characteristics, Normalized Difference Vegetation Index (NDVI), Land Use/Land Cover

Figure 6: SRTM Data Source (USGS Earth Explorer)

Applications of SRTM Data

- Terrain analysis
- Elevation mapping
- Flood risk assessment
- Geographic modeling

7. Combined Observations

- Bhuvan provides thematic satellite datasets for India
- SRTM provides elevation and terrain data
- Both datasets can be combined for comprehensive geospatial analysis

8. AI Pattern Recognition Applications

The datasets obtained from Bhuvan and SRTM can be used in AI/ML applications such as:

- Land use and land cover classification
- Vegetation and crop monitoring
- Water body detection
- Flood prediction using elevation data

Machine learning models, especially Convolutional Neural Networks (CNNs), can be used to identify patterns in satellite images and generate useful predictions.

9. Conclusion

Bhuvan and SRTM provide valuable geospatial datasets that are useful for environmental monitoring, research, and AI-based applications. These datasets enable analysis of spatial patterns and support decision-making in various domains.

10.Data Source Links

1. Bhuvan (ISRO) - Main Portal & Data Store

[1] Main Geoportal: https://bhuvan.nrsc.gov.in
[2] Bhuvan Store (For NDVI, Water, and Soil Data): https://bhuvan-app1.nrsc.gov.in/bhuvan2d/bhuvan/bhuvan2d.php
[3] Thematic Services (LULC&Maps): https://bhuvan-app1.nrsc.gov.in/thematic/thematic/index.php

2. SRTM & Elevation Data (USGS/NASA)

[4] USGS Earth Explorer (Primary Source for SRTM 30m): https://earthexplorer.usgs.gov
[5] NASA Earthdata(Alternative SRTM & Global Datasets) https://search.earthdata.nasa.gov